

Concepts Live in the Write Direction: Layered Evidence for Orthogonal, Composable LoRA Concept Spaces

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Abstract

Two low-rank adapters fine-tuned on disjoint domains (Spring Festival vs. summer) are shown to encode *separable concept spaces*: their weight updates ΔW are near-orthogonal (global overlap -0.002 to 0.135 across three seeds) while their input (read) subspaces are indistinguishable (all pairwise top- k similarities ≈ 3.8 , flat). A data-mixture “intermediate” adapter is an *interpolation*, not a third space (registered null model wins). Behavioural (lexical) separation is strong (hit rate 0.84 vs 0.11), yet output distributions of the two specialists are *closer to each other than either is to the base* at every position of an answer, on content and style tokens alike — composability is therefore a property of the weight geometry plus the gate, not of output distinguishability. A learned gate (0.90 accuracy; routed hit 0.800) beats a lexical router (0.80 ; 0.607), and the composed runtime $W = W_0 + \sum_i p_i \Delta W_i$ ships as 374–392 MB GGUF at 150–166 tokens/s on an M1 laptop. Cross-domain replication over five domains (adding code, legal, medical) keeps every specialist pair within $|\Delta W_{\text{ov}}| < 0.13$, with the interpolation anchor ($0.62/0.66$) proving the metric detects non-orthogonality when it exists. All numbers are pinned, four-way hash-verified, and reproducible from committed scripts; one self-audited sampling bug and its corrected rerun are reported in full.

1 Setup

Base: Qwen2.5-0.5B (frozen, manifest-verified bytes). Adapters: LoRA $r=16$, 7 target modules, lr 10^{-4} , 6 epochs, seeds 13/14/15; corpora of 55 (spring, summer), 110 (mixture), 40 (code/legal/medical, deviation registered) hand-written Chinese QA pairs. Probes: 40 questions (15/15/10), hashed *before* corpus authoring; 6-gram leakage tripwire = 0. Pre-registration precedes all runs (PREREG.md + four addenda); negative results are reported as first-class. Venue: chora/experiments/exp10-concept-adapters.

2 Method summary

Behaviour (10a): 5 sampled generations/probe; lexical hit rates; top-50 answer-start distributions, JS over union support. **Geometry (10b/10f):** per-module SVD of $A \in \mathbb{R}^{r \times d_{\text{in}}}$, $B \in \mathbb{R}^{d_{\text{out}} \times r}$; read-similarity $\|V_i V_j^\top\|_F$ at 90%-energy k , write-similarity $\|U_i^\top U_j\|_F$; basis-invariant global $\langle \Delta W_i, \Delta W_j \rangle / (\|\Delta W_i\| \|\Delta W_j\|)$; random-flat null $k/\sqrt{d}$; invasion = spectral band of the base weights the adapters write into. **Sequence (10c):** teacher-forced per-position JS on greedy reference texts; content-vs-style token classification under two definitions (function-word-inclusive, and punctuation-only). **Systems (10d/10g/GGUF):** multi-adapter runtime with

lexical router, then logistic gate on frozen-base sentence embeddings; merge, convert to GGUF f16, quantize q4_K_M, llama-bench tg128.

3 Results

pair	read V-A ($k90$)	write U-B ($k90$)	ΔW_{ov} (3 seeds)
spring–summer	3.818	0.55–1.24	−0.002...0.135
spring–mid	3.834	1.76–2.43	0.009...0.635
summer–mid	3.833	1.86–2.51	0.013...0.661
code/legal/medical cross pairs	3.785–3.790	1.07–1.16	≤ 0.129

Table 1: Orthogonal writes, shared reads; the mixture interpolates.

H1 fails with reversed sign (JS spring–summer 0.417/0.380 < base–arm 0.551/0.575); it survives full-sequence averaging but shrinks $3\times$ (gap 0.165 $\rightarrow$ 0.060) and remains smallest on *both* content and style classes under either definition — the plan’s content/style rescue is disconfirmed. **H2 passes** (0.84/0.49 vs base 0.11/0.33; cross ≤ 0.24 ; neutral ≈ 0). **H4 fails; registered null wins** (mixture = interpolation). **H6 passes:** max $|\Delta W_{ov}| = 0.1294$ across all 10 specialist pairs with positive control 0.618/0.661. **Invasion:** all adapters write the base’s *late* spectrum (band percentile ≈ 0.71 ; 1.5% energy in top-10% directions) — independent instrument, same finding as the workspace’s middle-eigenfunction thesis. **Systems:** gate 0.90 vs router 0.80 accuracy; routed hit 0.800 vs 0.607; q4_K_M 374–392 MB at $149.6 \pm 11.9/154.6 \pm 9.1/165.8 \pm 4.1$ t/s (target was 20).

4 Self-audit

The first router run reseeded *inside* the sampling loop: five “independent” draws were identical (reported 0.70 was a single-draw number in five costumes). Disclosed the same day, fixed, rerun (v2: 0.607); v1 bytes remain pinned and immutable. Two further first-run bugs (temperature polarity, negative concentration floor) were exposed by the companion park-side breathing experiment (C3) and fixed in both artefacts the same day.

5 Reading

Concept separation is where the adapter *writes*; what it *reads* is the shared commonsense of the base. $W_0 + \sum_i p_i \Delta W_i$ is therefore feasible as a composition law (near-orthogonality measured, five domains, one base size) *and* needs the gate: outputs alone do not reveal the concepts (10c), so the $\sum$ must be carried by a learned selector (10g). A time-modulated prior pointer is the selector’s natural extension — previewed symbolically in the sister park the same day.

6 Limits and next

One 0.5B base, toy corpora, one gate class; magnitudes not production claims. Exp11: 5 bases $\times$ 4 domains + universal-LoRA probe; MLP/neural controller; perplexity accounting for merged bf16.

Bytes: artifacts/results/exp10/ (13 pinned files, manifest four-way 0 failures); scripts train_lora.py · bench.py · analyze_10a.py · svd_10b.py · svd_seedcheck.py · kl_10c.py · kl_10c_grouped_S1.py · router_10d.py · gate_10g.py · gate_routed5.py · svd_10f.py · gguf_10e.sh; git: chora@a7a289e $\rightarrow$ e0e609f (this day’s receipts).